F-P-666D
December 31, 1973
SUPERSEDING
Int. Fed. Spec. F-P-00666C (GSA-FSS)
Notice 1, April 22, 1971,
Int. Fed. Spec. F-P-00666B (GSA-FSS)
January 4, 1963, and
Fed. Spec. F-P-666B
December 16, 1968

FEDERAL SPECIFICATION

PRIMER: PRESSURE, INTERNAL COMBUSTION ENGINE (COLD STARTING AID) AND PRIMER CARTRIDGES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers priming equipment for adding starting fluid to internal combustion engines as an aid to starting at low atmospheric temperatures. The primer is operated from a remote type control which is usually actuated from within the vehicle. The priming fuel is either contained under pressure in a cartridge or pressurized container, or in a container with actuating pump from which it is released to the engine through tubes, when actuated by the operator of the engine. Replacement cartridges or containers may be procured separately from the primer and operating mechanism by referencing this specification and specifying size, model and name of the manufacturer.

1.2 Classification.

1.2.1 Primers. The internal combustion engine cold-starting primers shall be of the following types and sizes as specified (see 6.2):

Type I - Power-pressurized cartridge and pressurized container.
Size 1 - 7.5 cu. cm.
Size 2 - 11 oz.
Size 3 - 20 oz.

Type II - Pump-operated.
Size 1 - 16 cu. cm.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Office, Washington, DC 20406.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods Of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publisher Officer., Room 1106, 222 South Rinsiac Plaza, Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Design. The primer shall aid starting of engines at low atmospheric temperatures by injecting low temperature starting fuels to create quicker combustion. It shall be designed to permit assembly with, and application to either gasoline or diesel engines. Unless otherwise specified (see 6.2), the starting aid equipment shall be of any type and size hereinafter specified and shall be actuated by either mechanical, electrical, hydraulic or pneumatic means. Primer shall be operable with one hand and when wearing arctic mittens. With the primer, container and priming fluid at -65°F., the primer shall dispense at least 95 percent of the priming fluid. The discharger shall be rugged and convenient to maintain. It shall be constructed in a manner to prevent escape of starting fluid into the atmosphere during and between starts, nor shall it permit leakage to the engine between starts. Nozzles shall be threaded to the standard 1/8-inch pipe thread size. The fluid container shall be held in the primer in such a manner as to hold the container securely so as to prevent damage to the primer or loosening of the container.

3.2 Performance. The engine primer and fluid shall start gasoline or diesel engines of any type and size at a temperature of minus 25°F. within one minute without application of heat. The starting shall be accomplished at a minimum cranking speed of 125 to 175 r.p.m. (depending upon the normal cranking speed of the vehicle),

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when provided with winter lubricant and fuels having a pour point at least 10 degrees lower than the ambient temperature of the vehicle. The fluid containers shall be capable of storage at any temperature from plus 155°F. for a 4-hour period to minus 65°F. for a period of 72 hours duration. The fluid containers shall not leak or rupture under these storage conditions (see 4.4.1). Gasket material shall be resilient at these temperatures.

3.3 Pressurized fluid container type primer. Primers shall consist of a device for holding containers (including tubing to the engine and atomizing nozzle), which will release the charge from the container and deliver it to the engine upon actuation by the operator. The primer shall be so arranged and constructed that it can be firmly installed in a location convenient to the equipment operator in normal engine-operating position. The arrangement and construction of the primer shall be such that efficient piercing or opening of the container and discharging the starting fluid to the engine will be accomplished. Unless otherwise specified, the primers shall be provided with a means to prevent discharge of starting fluid after the engine reaches normal operating temperature.

3.3.1 Type I, size 1. The discharger shall be a device for holding cartridges and piercing or rupturing the cartridges. It shall be possible to operate the dischargers with one hand.

3.3.2 Type I, size 2. The primer shall be of such design that it can be located outside the operator compartment near the engine to reduce the length of the starting fluid tubing, but controlled by the operator from his normal starting controls position. Two tubes with nozzles shall be furnished from the primer to the mainfolds when specifying for use on V-type dual manifolds, twin engines or split manifold engines. The size 2 primer shall be furnished with tubing to each manifold.

3.3.3 Type I, size 3. The primer shall be of such design that it can be located outside the operator compartment, but controlled by the operator from his normal operating seat. The primer valve shall have a safety sealing effect to hold the ether in case the valve in the container might fail to seat and leaks ether. Unless otherwise specified, the valve shall be so designed to measured amount of ether on each actuation of the primer by the operator. When specified for use on large engine applications, approximately 700 cubic inches or more (see 6.2), the valve shall be designed to continually release ether when the primer is actuated by the operator. Two tubes with nozzles shall be furnished from the primer to the manifolds when specifying for use on V-type dual manifolds, twin engines, or any split manifold engine. The primer shall be furnished with tubing leading to each manifold.

3.4 Type II, size 1, pump type primer. The type II, size 1, pump-type primer shall consist of a container containing starting fluid, and discharge equipment (including tubing to the engine) suitable for containing the container piercing or puncturing the container and releasing the charge and delivering it to the engine upon actuation of the pump by the operator. It shall be capable of being actuated by the operator from his normal engine starting position. The container shall be held in the primer in such a manner as to hold the container, securely so as to prevent damage to the primer or loosening of the container.

3.5 Cartridges and containers. All starting fluid cartridges and containers shall be in accordance with the Interstate Commerce Commission regulations, as applicable.

3.5.1 Type, size 1. The pressurized cartridge shall be of metal and of such design and construction that it shall withstand an internal pressure of 6,000 p.s.i. without rupture. The cartridge shall have 7.5 cu. cm. ether capacity. It external size shall be such that it snugly fits the cartridge chamber of the discharger (see 3.4.1). The cartridge shall be filled with a starting fluid of quantity, quality and characteristics to start the engine under the conditions of 3.2. After the cartridge is filled with starting fluid, it shall be pressurized with flammable gas. The internal equilibrium pressure of the cartridge at plus 70°F. shall be not less than 150 and not more than 325 p.s.i., and at minus 50°F. the internal pressure shall be not less than 50 p.s.i.

3.5.2 Type I, size 2. The container shall be of metal and equipped with a nozzle that will permit its use as a hand spray so that when holding the container in the hand, the operator can spray any desired amount of starting fluid into the air intake of the engine. The container shall have an 11 ounce capacity. The container

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shall also be of such design that it can, without tools, be inserted into and used in the type I, size 2 primer. The container shall be filled with a starting fluid of quantity, quality and characteristics to start the engine under the conditions of 3.2. After the container is filled with starting fluid, it shall be pressurized with flammable gas. The internal equilibrium pressure of the container at minus 65°F. shall be not less than 2 p.s.i. The container shall withstand an internal pressure of 300 p.s.i. without rupture.

3.5.3 Type I, size 3. The pressurized container shall be of metal and of such design and construction that it shall withstand an internal pressure of 850 p.s.i. without rupture. The container shall have a 20 ounce capacity. The container shall be filled with a starting fluid of quanitity, quality and characteristics to start the engine under the conditions of 3.2. After the container is filled with starting fluid, it shall be pressurized with flammable gas. The internal equilibrium pressure of the container at plus 70°F. shall be not less than 105 p.s.i., and at minus 50°F. the internal pressure shall be not less than 15 p.s.i. The container shall have a safety valve set for not less than 350 p.s.i.

3.5.4 Type II, size 1. Unless otherwise specified (see 6.2), the container shall be a metal container of a 16 cu. cm. capacity. Its external size shall be such that it snugly fits the container chamber of the discharger (see 3.6). The container shall be filled with starting fluid of quality, quanitity and characteristics to start the engine under the conditions of 3.2. The container shall withstand an internal pressure of 150 p.s.i. without rupture.

3.6 Starting fluid. The starting fluid used in the container shall be of such quantity and characteristics as to start the engine under the specified performance requirements of 3.2. The starting fluid shall contain a lubricant to serve as an upper cylinder lubricant and an inhibitor to prevent oxidation in the container. A color shall be included in the starting fluid of such composition and of sufficient quantity to effectively and visually indicate leakage of the charge from the container, even though the leak may occur above the liquid level.

3.7 Painting. Unless otherwise specified (see 6.2), the painting and finish of the primer equipment shall be the manufacturer's standard.

3.8 Spare parts. Spare parts as specified in the invitation to bid, contract or order shall be furnished with each unit (see 6.2).

3.9 Containers. Additional and replacement type containers may be procured by specifying this specification and manufacturer's name and model or size number of primer.

3.10 Instructions. Decal of instructions shall be placed on dash board of each unit supplied under this specification.

3.11 Workmanship. The unit shall be new and of the quality workmanship necessary to produce a unit of good appearance, having stability and efficient operating and productive characteristics as found in first class commercial quality items of a similar nature.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections and test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections and tests set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements (see 6.2).

4.2 Sampling.

4.2.1 Inspection lot. All primers of the same type offered for delivery at one time shall be considered a lot for purposes of sampling and inspection. All cartridges or containers of the same type and size offered for delivery at one time shall be considered a lot.

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4.2.2 Sampling for group A tests. From each lot of primers or cartridges. A random sample shall be selected in accordance with MIL-STD-105 inspection level II for the following tests:

Group A tests:
Visual and dimensional inspecton of primers, cartridges, or containers.
Capacity of cartridges or containers.
Leakage test.

If any primer or cartridge has one or more defects, the lot shall be rejected.

4.2.3 Sampling for group B tests. From each lot of primers, a random sample shall be selected in accordance with inspection level S-3 of the appendix to MIL-STD-105D dated March 20, 1964 for the following tests:

Group B tests:
Dispensing operation.

If any primer fails in any test, the lot represented shall be rejected.

4.2.4 Sampling for group C tests. From each procurement lot, sample primers shall be selected in accordance with inspection level L-2 of the appendix to MIL-STD-105D for the following test:

Group C test:
Engine starting at low temperature.

If any primer fails in this test, the cause of failure shall be determined and correction made in all apparatus delivered to the Government.

4.3 Inspection. Each sample primer selected in accordance with 4.2.2 shall be given a general inspection for material, workmanship, dimensions, performance weight, painting, marking, packaging, packing, paneling, and general compliance with the requirements of this specification.

4.4 Performance test. When applied to any engine, the primer shall start the engine under the conditions specified in 3.2.

4.4.1 Leak test. Cartridges and containers shall be tested for leaks under the conditions specified in 3.2.

4.4.2 Dispensing test. Cartridges and containers, priming fluid and primers shall all be stabilized at minus 65°F. and then discharged. At least 95 percent of the priming fluid shall be discharges (see 3.1).

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table 1. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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TABLE 1. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Number per container is more or less than specified. Net weight exceeds requirements of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packing shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each primer with cartridge, installation components, repair parts, maintenance tools, and technical publications shall be preserved together as a complete unit in accordance with MIL-P-116, method IA-15.

5.1.1.1 Additional cartridges. Cartridges of like description in the quantities specified (see 6.2), shall be preserved together in accordance with MIL-P-116, method IC-5.

5.1.2 Level C. Each primer with cartridge, installation components, repair parts, maintenance tools, and technical publications and additional cartridges, shall be preserved and packaged in a manner to afford protection against deterioration and damage. The supplier may use his standard commercial practice, providing that it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Primers of like description with cartridges, packaged as specified in 5.1, with additional cartridges of like description packaged as specified in 5.1 shall be packed together in quantities as specified (see 6.2) in a close-fitting box conforming to PPP-B-636, grade V3c. Close, waterproof seal and strap the boxes in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Primers and cartridges packaged as specified in 5.1 shall be packed as specified in 5.2.1 except that the boxes shall be class domestic and shall be sealed in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. Primers with cartridges and additional cartridges packaged as specified in 5.1.2 shall be packed to assure carrier acceptance and safe delivery at destination in containers complying with the National Motor Freight Classification rules or the Uniform Freight Classification rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be as specified in 5.3.1 or 5.3.2 (see 6.2).

5.3.1 Civil agencies. All marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. All marking shall be in accordance with MIL-STD-129.

6. NOTES

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6.1 Intended use. In general, type I and II primers are required to perform equal service in starting according to manufacturers recommendation.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of primer desired (see 1.2.1).
(c) Size of primer cartridges desired (see 1.2.1).
(d) If different, specify specific method of actuation desired; i.e., mechanical, electrical, hydraulic, or pneumatic means (see 3.1).
(e) If continued release primer valve is required (see 3.3.3).
(f) If other than 16 cu. cm. container is required (see 3.5.4).
(g) Color of paint, if different (see 3.7).
(h) Spare parts, when required (see 3.8).
(i) For additional cartridge procurement, state model and/or size, and name of manufacturer of primer (see 1.1 and 3.9).
(j) Place of inspection, if at contractor's plant (see 4.1).
(k) Level of preparation, packaging and packing required (see section 5).
(l) Marking, if different (see 5.3).

6.3 Data with bids. Each bidder should be required to furnish with his bid, complete descriptive data (drawings, specifications, handbook of instructions, etc.) covering the items he proposes to furnish under this specification and the invitation to bid.

6.4 Guarantee. It is suggested that the purchasing officer require the contractor to guarantee the equipment for a period of one year from the date of placing the equipment in regular operation against defective material or workmanship, and further to replace at point installation if installation is within the continental United States, or f.o.b. the continental United States port to be designated by the purchasing officer if installation is outside of the United States. Cost of installation shall be borne by the Government.

MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT: FIS, MVP
NASA: JFK

PREPARING ACTIVITY: GSA-FSS

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